

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester V Examination, 2022-23

Paper No. MTB-506: Operations Research (2012)

Time: 3 Hours

Full Marks: 50

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1.

- (a) Solve graphically: Maximize $3x + 4y$ subject to $x + y \leq 1$, $x \geq y$, $x, y \geq 0$. [2]
- (b) Find a basic feasible solution of: Minimize $[1, -2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ subject to $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. [2]
- (c) Find an extreme point of $X = \{(x, y) : x + y \leq 1, 0 \leq x, y \leq 1\}$. [2]
- (d) Consider a standard form LPP with $A = \begin{bmatrix} 0 & 2 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix}$, $b = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$, $c = \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}$. Suppose the reduced cost coefficient vector for some basis is $arc^T = [0, 1, 0, 0]$. Find an optimal solution to minimize the problem. [2]
- (e) Using part (d), find the value of the second cost coefficient a . [2]
- (f) Write the dual of: Minimize x subject to $x \leq 1$. [2]
- (g) Prove that the set of all optimal solutions of an LPP is a convex set. Illustrate with an example. [2]

Question 2.

- (a) State and prove the Fundamental Theorem of Linear Programming. [7]
- (b) Consider $Ax = b$ where $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & 0 & 2 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Find a (non-basic) feasible solution of $Ax = b$ and use the method in part (a) to find a basic feasible solution. [7]

Question 3.

- (a) Let X be the convex set of all feasible solutions of $Ax = b$, $x \geq 0$ where $A \in \mathbb{R}^{m \times n}$, $m < n$. Show that x is an extreme point of X if and only if $Ax = b$, $x \geq 0$ with the non-zero components of x forming linearly independent columns of A . [7]
- (b) Using the simplex method, solve: Maximize $7x + 6y$ subject to $2x + y \leq 3$, $x + 4y \leq 4$, $x, y \geq 0$. [7]
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Question 4.

- (a) Show that the original LPP has a basic feasible solution if and only if the associated artificial problem has an optimal feasible solution with objective function value zero. [6]
- (b) Solve: Minimize $x_1 + 2x_2$ subject to $x_1 + x_2 \geq 25$, $-x_1 + 2x_2 \geq 3$, $2x_1 + x_2 \leq 4$, $x_1, x_2 \geq 0$. [8]
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Question 5.

- (a) If the primal problem (in either form) has an optimal solution, show that the dual also has an optimal solution and the optimal values of the objective functions are equal. [7]
- (b) Use the dual simplex algorithm to solve: Minimize $x_1 + x_2 + x_3$ subject to $x_1 + x_2 + x_3 \geq 3$, $4x_1 + x_2 + 2x_3 \geq 5$, $x_1, x_2, x_3 \geq 0$. [7]
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Question 6.

- (a) State and prove the Complementary Slackness Theorem in relation to primal and dual linear programming problems. [7]
- (b) You have 26 Euros and wish to purchase gold. Four vendors sell at prices 4, 1, 4, 4 Euros per ounce respectively. Using complementary slackness, find an optimal solution to purchase gold spending all 26 Euros. [7]
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Question 7.

- (a) Find an optimal solution of the transportation problem:

	D_1	D_2	D_3	D_4
Supply				
S_1	3	2	7	6
50				
S_2	7	5	2	3
60				
S_3	2	5	4	5
25				
Demand	60	40	20	15

[8]

(b) Solve the assignment problem using the Hungarian Algorithm to minimize total time:

	J_1	J_2	J_3	J_4
J_5				
P_1	6	5	4	6
5				
P_2	3	6	5	6
5				
P_3	5	6	5	3
7				
P_4	7	6	3	5
6				
P_5	5	7	6	8
6				

[6]